

● MEDIUM

● EXPRESSION OF IDEAS

# Mixed

10 questions · ~15 min

# 01

Q1

EXPRESSION OF IDEAS

Earth's auroras—colorful displays of light seen above the northern and southern poles—result, broadly speaking, from the Sun's activity. \_\_\_\_\_ blank the Sun releases charged particles that are captured by Earth's magnetic field and channeled toward the poles. These particles then collide with atoms in the atmosphere, causing the atoms to emit auroral light.

Which choice completes the text with the most logical transition?

- A. Specifically,
- B. Similarly,
- C. Nevertheless,
- D. Hence,

While researching a topic, a student has taken the following notes:

- Severo Ochoa discovered the enzyme PNPase in 1955.
- PNPase is involved in both the creation and degradation of mRNA.
- Ochoa incorrectly hypothesized that PNPase provides the genetic blueprints for mRNA.
- The discovery of PNPase proved critical to deciphering the human genetic code.
- Deciphering the genetic code has led to a better understanding of how genetic variations affect human health.

The student wants to emphasize the significance of Ochoa's discovery. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Ochoa's 1955 discovery of PNPase proved critical to deciphering the human genetic code, leading to a better understanding of how genetic variations affect human health.
- B. Ochoa first discovered PNPase, an enzyme that he hypothesized contained the genetic blueprints for mRNA, in 1955.
- C. In 1955, Ochoa discovered the PNPase enzyme, which is involved in both the creation and degradation of mRNA.
- D. Though his discovery of PNPase was critical to deciphering the human genetic code, Ochoa incorrectly hypothesized that the enzyme was the source of mRNA's genetic blueprints.

In his 1925 book *The Morphology of Landscape*, US geographer Carl Sauer challenged prevailing views about how natural landscapes influence human cultures.

\_\_\_\_\_blank Sauer argued that instead of being shaped entirely by their natural surroundings, cultures play an active role in their own development by virtue of their interactions with the environment.

Which choice completes the text with the most logical transition?

- A. Similarly,
- B. Finally,
- C. Therefore,
- D. Specifically,

Cats can judge unseen people's positions in space by the sound of their voices and thus react with surprise when the same person calls to them from two different locations in a short span of time. Saho Takagi and colleagues reached this conclusion by measuring cats' levels of surprise based on their ear and head movements while the cats heard recordings of their owners' voices from two speakers spaced far apart. Cats exhibited a low level of surprise when owners' voices were played twice from the same speaker, but they showed a high level of surprise when the voice was played once each from the two different speakers.

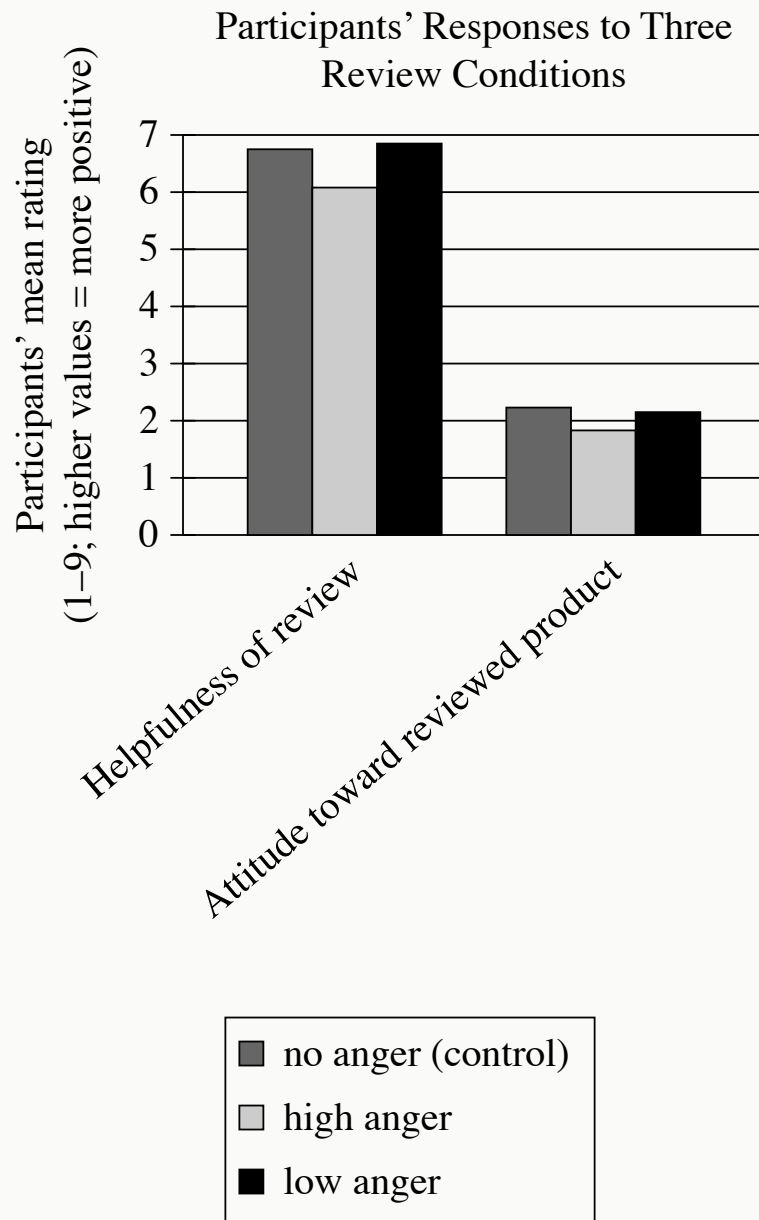
According to the text, how did the researchers determine the level of surprise displayed by the cats in the study?

- A. They watched how each cat moved its ears and head.
- B. They examined how each cat reacted to the voice of a stranger.
- C. They studied how each cat physically interacted with its owner.
- D. They tracked how each cat moved around the room.

Among the most visited art museums in the world, the Galleria dell'Accademia in Florence had approximately 1.7 million visitors in 2019. The Galleria dell'Accademia also offers virtual tours that art lovers can view online for free. Although there were initial concerns that people who viewed the virtual tours would then consider an in-person visit unnecessary, museum administrators claim that their surveys of in-person visitors show that those concerns were unjustified.

Which statement, if true, would most directly support the administrators' claim?

- A. Many surveyed visitors to the Galleria dell'Accademia indicated that the virtual tours convinced them to plan an in-person visit.
- B. Most surveyed visitors to the Galleria dell'Accademia indicated that they were unaware of the virtual tours before their first in-person visit.
- C. Most surveyed visitors to the Galleria dell'Accademia indicated that they lived somewhere other than Florence.
- D. Many surveyed visitors to the Galleria dell'Accademia indicated that they would likely view the virtual tours in order to reminisce about their in-person visit.



- For each data category, the following bars are shown:
  - no anger (control)
  - high anger
  - low anger
- The data for the 2 categories are as follows:
  - Helpfulness of review:

- no anger (control): 6.75
- high anger: 6.08
- low anger: 6.85
- Attitude toward reviewed product:
  - no anger (control): 2.23
  - high anger: 1.83
  - low anger: 2.15

To understand how expressions of anger in reviews of products affect readers of those reviews, business scholar Dezhi Yin and colleagues measured study participants' responses to three versions of the same negative review—a control review expressing no anger, a review expressing a high degree of anger, and a review expressing a low degree of anger. Reviewing the data, a student concludes that the mere presence of anger in a review may not negatively affect readers' perceptions of the review, but a high degree of anger in a review does worsen readers' perceptions of the review.

Which choice best describes data from the graph that support the students' conclusion?

- A. On average, participants' ratings of the helpfulness of the review were substantially higher than were participants' ratings of the reviewed product regardless of which type of review participants had seen.
- B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.
- C. Participants who saw the low-anger review rated the review as slightly more helpful than participants who saw the control review did, but participants' attitude toward the reviewed product was slightly worse when participants saw the low-anger review than when they saw the no-anger review.
- D. Compared with participants who saw the low-anger review, participants who saw the high-anger review rated the review as less helpful and had a less positive attitude toward the reviewed product.

Cy Twombly, a US painter and sculptor, created many large-scale abstract works, such as his 10-painting series *Fifty Days at Iliam*. In these works, Twombly's artistic style is exemplified by his use of graffiti-like \_\_\_\_\_ blank often incorporate words or phrases from poetry and mythology.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. scribbles: that
- B. scribbles that
- C. scribbles; that
- D. scribbles. That

Far from being modern inventions, \_\_\_\_\_blank more than 5,000 years ago.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. Sumerians in ancient Mesopotamia used drinking straws
- B. drinking straws were used by Sumerians in ancient Mesopotamia
- C. the use of drinking straws by Sumerians in ancient Mesopotamia happened
- D. ancient Mesopotamia was home to Sumerians who used drinking straws

**Text 1**

Little is known about how plate tectonics—wherein slabs of Earth’s crust move over, under, away from, and against one another—began. Some researchers contend that tectonic movements began around 3 billion years ago, often noting that computer models of Earth’s mantle temperature at the time indicate that the mantle would have been sufficiently molten to enable the plates to move.

**Text 2**

Ultimately, any plausible claim about the inception of tectonic movement must rest on empirical evidence from the geological record. Researcher Wriju Chowdhury and his team analyzed the geochemistry of zircon crystals to gain insight into the chemical composition of the magma from which the crystals formed and, based on the data, compellingly argue that plate tectonics may have been occurring as early as 4.2 billion years ago.

Based on the texts, how would the author of Text 2 most likely respond to what “some researchers contend” as described in Text 1?

- A. By suggesting that the temperature of Earth’s mantle 3 billion years ago was likely insufficient to allow for the level of tectonic movement predicted by computer models
- B. By distinguishing between computer models of Earth’s mantle temperature that reliably predict the onset of plate tectonics and those that do not
- C. By indicating that computer models of Earth’s mantle temperature are still being improved such that new models tend to be much more reliable than their predecessors

- D. By asserting that a more definitive form of evidence than the computer models suggests a different timeline for the onset of plate tectonics on Earth

**Q10**

CRAFT AND STRUCTURE

Economists often assert that countries looking to increase their reliance on solar energy should expand their capacity for storage; having an ample reserve of stored energy can mitigate the effects of \_\_\_\_\_blank solar energy collection caused by unpredictable shifts in cloud cover and haze.

Which choice completes the text with the most logical and precise word or phrase?

- A. developments of
- B. fluctuations in
- C. calibrations with
- D. incentives for